

Date: 2026-05-13

Request ID: 100048533

Hotplate		
Element(s)	Na	
SOP#		
Cocktail	60 mL HCl	
sample(s)	weight (g)	volume (mL)
200127922_1	0.1564	250
200127922_2	0.1577	250
200127923_1	0.1543	250
200127923_2	0.1572	250

Date: 2026-05-13
Request ID: 100048533

LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200127922	27.1548	29.1688	29.1604	0.42	0.9958

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm [dried]	%Na [dried]
200127922_1	250/250	0.645	1035.33	0.10
200127922_2	250/250	0.645	1026.79	0.10

na result: 1031.06 ppm

na result: 0.10 %

na oxide factor: 1.348

na oxide result: 0.14 %

na lot: *Manufacturer: Assurance, 27-159NAY exp: 2026-08-30*

Note(s):

Date: 2026-05-13
Request ID: 100048533

LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200127923	26.4679	28.4893	28.4773	0.59	0.9941

Na ICP analysis

sample	Dilution	conc. [mg/L]	Na_ppm [dried]	%Na [dried]
200127923_1	250/250	0.720	1173.53	0.12
200127923_2	250/250	0.720	1151.88	0.12

na result: 1162.70 ppm

na result: 0.12 %

na oxide factor: 1.348

na oxide result: 0.16 %

na lot: *Manufacturer: Assurance, 27-159NAY exp: 2026-08-30*

Note(s):

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$ppm\ M+ = [conc.][volume][dilution]/[weight]$$

$$\%M+ = [conc.][volume][dilution]/([weight]*10,000)$$

$$\%MO = [oxide\ factor]*\%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1-(\%LOI)/100])$$

$$\%M+ \ [dried] = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \ [dried] = \%MO / ([1-(\%LOI)/100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS])/([weight])$$

$$\%Cr_2O_3 = 1.462*[\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$